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S **COMMON HALF YEARLY EXAMINATION - 2025**

Standard - 12 Reg. No. | | | | | | |

Time Allowed:3.00 Hours **MATHEMATICS** *Maximum Marks:90*

PART-I

I. Answer all the questions: **20×1=20**

1) If A^TA^{-1} is symmetric, then A^2 =

a) A^{-1} b) $(A^T)^2$ c) A^T d) $(A^{-1})^2$

2) If $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$ be such that $\lambda A^{-1} = A$, then λ is

a) 17 b) 14 c) 19 d) 21

3) The principal argument of $(\sin 40° + i \cos 40°)^5$ is

a) $-110°$ b) $-70°$ c) $70°$ d) $110°$

4) If $\left|z - \frac{3}{2}\right| = 2$ then the least value of $|z|$ is

a) 1 b) 2 c) 3 d) 5

5) If α, β and γ are the zeros of $x^3 + px^2 + qx + r$, then $\sum \frac{1}{\alpha}$ is

a) $\frac{q}{r}$ b) $\frac{-p}{r}$ c) $\frac{q}{r}$ d) $\frac{-q}{p}$

6) If $\cot^{-1}(\sqrt{\sin\alpha}) + \tan^{-1}(\sqrt{\sin\alpha}) = u$, then cos 2u is equal to

a) $\tan^2\alpha$ b) 0 c) -1 d) tan2 α

7) If $x + y = k$ is a normal to the parabola $y^2 = 12x$, then the value of k is

a) 3 b) -1 c) 1 d) 9

8) If the direction cosines of a line are $\frac{1}{C}, \frac{1}{C}, \frac{1}{C}$ then

a) $C = \pm 3$ b) $C = \pm\sqrt{3}$ c) $C > 0$ d) $0 < C < 1$

9) The slope of the line normal to the curve $f(x) = 2\cos 4x$ at $x = \pi/12$ is

a) $-4\sqrt{3}$ b) -4 c) $\frac{\sqrt{3}}{12}$ d) $4\sqrt{3}$

10) The maximum value of the function $x^2 e^{-2x}$, $x > 0$ is

a) $\frac{1}{e}$ b) $\frac{1}{2e}$ c) $\frac{1}{e^2}$ d) $\frac{4}{e^4}$

11) If $w(x,y) = x^y$, $x > 0$, then $\frac{\partial w}{\partial x}$ is equal to

a) $x^y \log x$ b) $y \log x$ c) yx^{y-1} d) $x \log y$

12) The value of $\int_0^1 x(1-x)^{99}\,dx$ is

a) $\frac{1}{11000}$ b) $\frac{1}{10100}$ c) $\frac{1}{10010}$ d) $\frac{1}{10001}$

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13) The value of $\int_0^a \left(\sqrt{a^2 - x^2}\right)^3 dx$ is

a) $\frac{\pi a^3}{16}$ b) $\frac{3\pi a^4}{16}$ c) $\frac{3\pi a^2}{8}$ d) $\frac{3\pi a^4}{8}$

14) The integrating factor of the differential equation $\frac{dy}{dx} + y = \frac{1+y}{x}$ is

a) $\frac{x}{e^x}$ b) $\frac{e^x}{x}$ c) λe^x d) e^x

15) If P(X=0) = 1 – P(X=1). If E(X) = 3 Var(X) then P(X=0) is

a) $\frac{2}{3}$ b) $\frac{2}{5}$ c) $\frac{1}{5}$ d) $\frac{1}{3}$

16) The operation * defined by $a*b = \frac{ab}{7}$ is not binary operation on

a) Q^+ b) Z c) R d) C

17) A random variable X is a function from

a) $S \to R$ b) $R \to S$ c) $S \to N$ d) $N \to S$

18) The value of $(\vec{a} - \vec{b}) \times (\vec{a} + \vec{b})$

a) $a^2 - b^2$ b) $2(\vec{a} \times \vec{b})$ c) $\vec{a} \times \vec{b}$ d) $\vec{b} \times \vec{a}$

19) The conic section for the equation e^{x^3+y} $x^2 - 3 = 5x + 3y$ is

a) hyperbola b) an ellipse c) a circle d) a parabola

20) The order and degree of the differential equation $x = e^{xy}\left(\frac{dy}{dx}\right)$ are

a) (1,1) b) (0,1) c) (1,0) d) (2,1)

PART-II

Answer any seven questions: **7×2=14**

Question No.30 is compulsory.

21) Verify the property $(A^T)^{-1} = (A^{-1})^T$ with $A = \begin{bmatrix} 2 & 9 \\ 1 & 7 \end{bmatrix}$.

22) Find the square root of $-6 + 8i$.

23) If α, β and γ are the roots of the equation $x^3 + px^2 + qx + r = 0$, find the value of $\Sigma \frac{1}{\beta\gamma}$ in terms of the coefficient.

24) Find the value of $\sin^{-1}\left(\sin \frac{2\pi}{3}\right)$.

25) Show that the vectors $2\vec{i} - \vec{j} + 3\vec{k}$, $\vec{i} - \vec{j}$ and $3\vec{i} - \vec{j} + 6\vec{k}$ are coplanar.

26) Find df for $f(x) = x^2 + 3x$ and evaluate it for x = 3 and dx = 0.02.

27) From the differential equation by eliminating the arbitary constant A and B from $y = A \cos x + B \sin x$.

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28) If X is a random variable with distribution function F(x) given by,

$$F(x) = \begin{cases} 0 & ; \ -\infty < x < 0 \\ \frac{1}{2}(x^2 + x) & ; \ 0 \le x < 1 \\ 1 & ; \ 1 \le x < \infty \end{cases}$$

then find the probability density function f(x).

29) Let $A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ be any two Boolean matrices of the same type. Find $A \vee B$ and $A \wedge B$.

30) Find all values of C in the interval (−2,2) such that f'(C) = 0, for the function $f(x) = x^4 - 2x^2$.

PART-III

Answer any seven questions: **7×3=21**
Question No.40 is compulsory.

31) Find the rank of the matrix $\begin{bmatrix} 1 & 1 & 1 & 3 \\ 2 & -1 & 3 & 4 \\ 5 & -1 & 7 & 11 \end{bmatrix}$.

32) The complex numbers u, v and w are related by $\frac{1}{u} = \frac{1}{v} + \frac{1}{w}$ if v = 3 − 4i and w = 4 + 3i, find u in rectangular form.

33) Find the value of $\cos^{-1}\left(\cos\frac{\pi}{7}\cos\frac{\pi}{17} - \sin\frac{\pi}{7}\sin\frac{\pi}{17}\right)$.

34) The maximum and minimum distance of the Earth from the Sun respectively are 152×10^6 km and 94.5×10^6 km. The Sun is at one focus of the elliptical orbit. Find the distance from the Sun to the other focus.

35) Find the torque of the resultant of the three forces represented by $-3\vec{i} + 6\vec{j} - 3\vec{k}$, $4\vec{i} - 10\vec{j} + 12\vec{k}$ and $4\vec{i} + 7\vec{j}$ acting at the point with position vector $8\vec{i} - 6\vec{j} - 4\vec{k}$, about the point with position vector $18\vec{i} + 3\vec{j} - 9\vec{k}$.

36) Evaluate : $\lim_{x \to 0^+} x \log x$.

37) Use the linear approximation to find the approximation value of $(123)^{\frac{2}{3}}$.

38) Evaluate : $\int_0^{\pi/3} \frac{\sec x \tan x}{1 + \sec^2 x} dx$.

39) Prove that $q \to p \equiv \neg p \to \neg q$.

40) Solve $\frac{dy}{dx} = e^{x+y} + x^2 e^{x^3+y}$.

PART-IV

Answer all questions: **7×5=35**

41) a) $x - y + 2z = 2$; $2x + y + 4z = 7$; $4x - y + z = 4$ test for consistency and it possible, solve the system of equation by rank method. **(OR)**

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b) A police jeep, approaching an orthogonal intersection from the northern direction, is chasing a speeding car that has turned and moving straight east. When the Jeep is 0.6 km north of the intersection and the car is 0.8 km to the east. The police determine with a radar that the distance between them and the car is increasing at 20 km/hr. If the jeep is moving at 60 km/hr at the instant of measurement, what is the speed of the car?

42) a) If $z = x + iy$ and $\arg\left(\frac{z-i}{z+2}\right) = \frac{\pi}{4}$ then show that $x^2 + y^2 + 3x - 3y + 2 = 0$.

(OR)

b) Find the area of the region bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

43) a) If $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi$ show that $x + y + z = xyz$. **(OR)**

b) On lighting a rocket cracker it gets projected in a parabolic path and reaches a maximum height of 4m when it is 6m away from the point of projection. Finally it reaches the ground 12m away from the starting point. Find the angle of projection.

44) a) Using Vector method, prove that $\cos(\alpha - \beta) = \cos\alpha \cos\beta + \sin\alpha \sin\beta$.

(OR)

b) Find the dimensions of the largest rectangle that can be inscribed in a semi circle of radius r cm.

45) a) A six sided die is marked '1' on one face, '3' on two of its faces and '5' on remaining three faces. The die is thrown twice. If X denote the total score in two throws, find (i) the probability mass function (ii) the cumulative distribution function (iii) $P(4 \le X \le 10)$ (iv) $P(X \ge 6)$. **(OR)**

b) If $V(x,y) = \log\left(\frac{x^2+y^2}{x+y}\right)$ prove that $x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y} = 1$.

46) a) Find the population of a city at any time t, given that the rate of increase of population is proportional to the population at that instant and that in a period of 40 years the population increased from 3,00,000 to 4,00,000. **(OR)**

b) Let $M = \left\{\begin{pmatrix} x & x \\ x & x \end{pmatrix} ; x \in R - \{0\}\right\}$ and let * be the matrix multiplication. Determine whether M is closed under *. If so examine the commutative and associative properties satisfied * on M.

47) a) Find the parametric form of Vector equation and Cartesian equation of the plane containing the line $\vec{r} = \left(\vec{i} - \vec{j} + 3\vec{k}\right) + t\left(2\vec{i} - \vec{j} + 4\vec{k}\right)$ and perpendicular to plane $\vec{r} \cdot \left(\vec{i} + 2\vec{j} + \vec{k}\right) = 8$. **(OR)**

b) Evaluate $\int_2^3 \frac{\sqrt{x}}{\sqrt{5-x} + \sqrt{x}}\, dx$.

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